

PROJECT REPORT : COCOMO MODEL **APPLICATION**

Introduction

This report details the application of the COCOMO (Constructive Cost Model) for software project estimation for the KuCampusBite application. The report outlines key project metrics, including size, effort, duration, and required staffing, and provides a breakdown of the cost drivers used in the estimation process.

Project Summary

- **Project Name:** KuCampusBite
- **Project Type:** Organic
- **Project Size:** 8.05 KLOC (thousand lines of code)
- **Effort Adjustment Factor (EAF):** 1.53

Project Metrics

The following metrics were derived from the COCOMO model to estimate project requirements:

- **Project Effort:** 33 person-months

Calculation: $1.53 \times 2.4 \times 8.051.05 = 32.8$ person-months

- **Project Duration:** 9.5 months

Calculation: $2.5 \times 32.80.38 = 9.41$ months

- **Project Staff:** 4 persons

Calculation: $33 \text{ person-months} / 9.5 \text{ months} = 3.5$ persons

Cocomo Cost Drivers And Multipliers

No#	Cost Driver	Category	Rating	Effort M (EM)
1	RELY (Required Software Reliability)	Product	Nominal	1.00
2	DATA (Database Size)	Product	Nominal	1.00
3	CPLX (Product Complexity)	Product	Nominal	1.00
4	Time (Execution Time Constraint)	Hardware	Nominal	1.00
5	STOR (Storage Constraint)	Hardware	Nominal	1.00
6	VIRT (Platform Volatility)	Hardware	Nominal	1.00
7	TURN (Turnaround Time)	Hardware	Nominal	1.00
8	ACAP (Analyst Capability)	Personnel	Low	1.19
9	PCAP (Programmer Capability)	Personnel	Low - borderline Nominal	1.17
10	AEXP (Application Experience)	Personnel	Very Low	1.29
11	VEXP (Platform Experience)	Personnel	Nominal	1.00
12	LEXP (Language/ Tool Experience)	Personnel	Nominal	1.00
13	TOOL (Use of Software Tools)	Project	Nominal	1.00
14	MODP (Modern Programming Practice)	Project	Very High	0.82
15	SCED (Schedule Constraint)	Project	High	1.04

